

LUCIANO BONADONA

(321) 945-7467 • lbsiles01@gmail.com • Fayetteville, AR 72701

github.com/lbonadonasiles-lgtm • lbonadonasiles-lgtm.github.io • linkedin.com/in/luciano-bonadona

PROFESSIONAL SUMMARY

I am a recent Biomedical Engineering graduate who spent my academic career building things, from engineering projects that impacted people's lives to ML pipelines, ETL workflows, SQL databases, and dashboards across healthcare, energy, and neurotech. I work in Python, SQL, and R while keeping everything on GitHub. I turn messy data into clear stories for people who don't live in spreadsheets. Trilingual in English, Spanish, and French.

CORE COMPETENCIES

SQL & ETL Pipelines • Data Modeling • Python (pandas, NumPy, scikit-learn) • Machine Learning • Statistical Analysis • Hypothesis Testing • Regression & Classification • Feature Engineering • KPI Development • Data Architecture • Exploratory Data Analysis • Data Visualization • A/B Testing Concepts • Git / GitHub • Actionable Insight Generation • Cross-Functional Communication

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, XGBoost), SQL, R, C++

ML & Analytics: Machine learning (classification, regression, clustering), statistical analysis, hypothesis testing, feature engineering, EDA, predictive modeling, data pipeline development

Data Engineering: ETL pipelines, relational database design (SQL/SQLite), star-schema data modeling, data quality assessment, data governance, data integrity

Visualization & Reporting: Power BI, Excel (advanced), KPI dashboards, stakeholder reporting, ad hoc analysis

Tools & Version Control: Git/GitHub, Jupyter Notebook, AWS & Azure (basics), MATLAB, SolidWorks | CSWA, CSWA-S, First Aid/CPR/AED

Languages: Fluent in English, Spanish, and French

EDUCATION

B.S. in Biomedical Engineering, Minor in Computer Science - University of Arkansas

May 2026

Fayetteville, AR | GPA: 3.4

Relevant Coursework: Biostatistics, Data Structures, Linear Algebra, Signals and Systems, Modeling, Productivity Improvement (Lean)

PROJECTS

Energy Demand Forecasting - Python (pandas, scikit-learn), XGBoost

Jan 2026 - Present

- Built an end-to-end ML pipeline on 2M+ time-series data points; defined and engineered lag and time-based features, improving model R2 from 0.30 to 0.61; benchmarked Linear Regression, Random Forest, and XGBoost to identify best-performing model and documented findings on GitHub.
- Applied hypothesis-driven feature selection: tested seasonal and hourly decomposition hypotheses, validated results against holdout data, and reported model performance using RMSE, MAE, and R2 metrics.

EEG Seizure Detection - Python (scikit-learn, XGBoost)

2026

- Engineered 12 statistical features from raw physiological time-series (11,500 samples); trained and compared classification models (Random Forest, XGBoost) achieving 98% accuracy; evaluated precision, recall, F1, and confusion matrix to assess real-world deployment risk; all methodology version-controlled on GitHub.

Healthcare Readmission Analytics - SQL, Python (pandas), Power BI

2026

- Designed a normalized SQL star-schema relational database from 25,000 patient encounters with enforced key integrity; built an ETL pipeline covering ingestion, EDA, data quality checks, and transformation.
- Identified prior inpatient utilization as the strongest readmission driver (rates 40% to 81%) through structured exploratory analysis; developed KPI dashboard in Power BI to communicate findings and recommendations to non-technical stakeholders.

Go Baby Go! - Adaptive Mobility Device - University of Arkansas

Fall 2025 - May 2026

- Led a team of 5 engineers to adapt a ride-on vehicle for a child with limited lower-limb motor control; collaborated with clinicians to translate clinical mobility requirements into engineering specifications.
- Rewired the vehicle's battery system to redirect power from the foot-pedal actuator to a custom push-button accelerator, enabling independent, safe mobility without requiring lower-limb use.

WORK EXPERIENCE

Process Optimization & Analytics Intern - eSoft S.R.L.

May-Aug 2024, May-Aug 2025

La Paz, Bolivia

- Analyzed operational data workflows and delivered data-driven recommendations reducing processing time by 20-30%; identified root-cause inefficiencies through structured data analysis and validated improvements against baseline KPIs.
- Built Power BI and Excel dashboards tracking KPI trends (turnaround time, delays, workflow performance), enabling real-time visibility and self-service reporting for cross-functional stakeholders.
- Documented data requirements and designed rule-based automation logic; partnered with vendors and technical teams throughout build and validation.

Lifeguard - University of Arkansas

May 2022 - Present

Fayetteville, AR

- Enforce safety and compliance standards for a facility serving 100+ daily users; maintain precise incident documentation.

VOLUNTEERING

Camp Counselor, Camp Frontier - Tampa, FL (Summer 2022) | **Recreation Aide**, Anne Kolb Nature Center - Hollywood, FL (Summer 2023)